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Impact of artificial intelligence on human loss in decision making, laziness and safety in education

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This study examines the impact of artificial intelligence (AI) on loss in decision making, laziness, and privacy concerns among university students in Pakistan and China. Like other sectors, education also adopts AI technologies to address modern-day challenges. AI investment will grow to USD 253.82 million from 2021 to 2025. However, worryingly, researchers and institutions across the globe are praising the positive role of AI but ignoring its concerns. This study is based on qualitative methodology using PLS Smart for the data analysis. Primary data was collected from 285 students from different universities in Pakistan and China. The purposive Sampling technique was used to draw the sample from the population. The data analysis findings show that AI significantly impacts the loss of human decision-making and makes humans lazy. It also impacts security and privacy. The findings show that 68.9% of laziness in humans, 68.6% in personal privacy and security issues, and 27.7% in the loss of decision-making are due to the impact of artificial intelligence in Pakistani and Chinese society. From this, it was observed that human laziness is the most affected area due to AI. However, this study argues that significant preventive measures are necessary before implementing AI technology in education. Accepting AI without addressing the major human concerns would be like summoning the devils. Concentrating on justified designing and deploying and using AI for education is recommended to address the issue.

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Introduction

Artificial intelligence (AI) is a vast technology used in the education sector. Several types of AI technology are used in education (Nemorin et al., 2022). Majorly includes Plagiarism Detection, Exam Integrity (Ade-Ibijola et al., 2022), Chatbots for Enrollment and Retention (Nakitare and Otike, 2022), Learning Management Systems, Transcription of Faculty Lectures, Enhanced Online Discussion Boards, Analyzing Student Success Metrics, and Academic Research (Nakitare and Otike, 2022). Nowadays, Education Technology (EdTech) companies are deploying emotional AI to quantify social and emotional learning (McStay, 2020). Artificial intelligence, affective computing methods, and machine learning are collectively called "emotional AI" (AI). Artificial intelligence (AI) shapes our future more powerfully than any other century's invention. Anyone who does not understand it will soon feel left behind, waking up in a world full of technology that feels more and more like magic (Maini and Sabri, 2017). Undoubtedly, AI technology has significant importance, and its role has been witnessed in the recent pandemic. Many researchers agree it can be essential in education (Sayed et al., 2021). but this does not mean it will always be beneficial and free from ethical concerns (Dastin, 2018). Due to this, many researchers focus on its development and use but keep their ethical considerations in mind (Justin and Mizuko, 2017). Some believe that although the intentions behind AI in education may be positive, this may not be sufficient to prove it ethical (Whittaker and Crawford, 2018).

There is a severe need to understand the meaning of being "ethical" in the context of AI and education. It is also essential to find out the possible unintended consequences of the use of AI in education and the main concerns of AI in education, and other considerations. Generally, AI's ethical issues and concerns are innovation cost, consent issues, personal data misuse, criminal and malicious use, freedom and autonomy loss, and the decision making loss of humans, etc. (Stahl B. C., 2021a, 2021b). Although, technology also enhances organizational information security (Ahmad et al., 2021) and competitive advantage (Sayed and Muhammad, 2015) and enhances customer relationships (Rasheed et al., 2015). Researchers are afraid that by 2030 the AI revolution will focus on enhancing benefits and social control but will also raise ethical concerns, and there is no consensus among them. A clear division regarding AI's positive impact on life and moral standing (Rainie et al., 2021).

It is evident from the literature on the ethics of AI that besides its enormous advantages, many challenges also emerge with the development of AI in the context of moral values, behavior, trust, and privacy, to name a few. The education sector faces many ethical challenges while implementing or using AI. Many researchers are exploring the area further. We divide AI in education into three levels. First, the technology itself, its manufacturer, developer, etc. The second is its impact on the teacher, and the third is on the learner or student.

Foremost, there is a need to develop AI technology for education, which cannot be the basis of ethical issues or concerns (Ayling and Chapman, 2022). The high expectations of AI have triggered worldwide interest and concern, generating 400+ policy documents on responsible AI. Intense discussions over ethical issues lay a helpful foundation, preparing researchers, managers, policymakers, and educators for constructive discussions that will lead to clear recommendations for building reliable, safe, and trustworthy systems that will be a commercial success (Landwehr, 2015). But the question is, is it possible to develop an AI technology for education that will never cause an ethical concern? Maybe the developer or the manufacturer has dishonest gain from the AI technology in education. Maybe their intentions are not towards the betterment and assistance of education. Such questions come to mind when someone talks about the impact of AI in Education. Even if the development of AI technology is clear from any ethical concerns from the developer or manufacturer, there is no guarantee for the opposite view. The risk of ethical considerations will also rely upon the technical quality. Higher quality will minimize the risk but is it possible for all educational institutions to implement expensive technology of higher quality? (Shneiderman, 2021). Secondly, many issues may arise when teachers use AI technology (Topcu and Zuck, 2020). It may be security, usage, implementation, etc. Questions about security, bias, affordability, trust, etc., come to mind (IEEE, 2019). Thirdly, privacy, trust, safety, and health issues exist at the user level. To address such questions, a robust regulatory framework and policies are required. Still, unfortunately, no framework has been devised, no guidelines have been agreed upon, no policies have been developed, and no regulations have been enacted to address the ethical issues raised by AI in education (José et al., 2018).

It is evident that AI technology has many concerns (Stahl B. C., 2021a, 2021b), and like other sectors, the education sector is also facing challenges (Hax, 2018). If not all the issues/problems directly affect education and learning, most directly or indirectly impact the education process. So, it is difficult to decide whether AI has a positive ethical impact on education or negative or somewhat positive or negative. The debate on ethical concerns about AI technology will continue from case to case and context to context (Petousi and Sifaki, 2020). This research is focused on the following three moral fears of AI in education:

1. Security and privacy
2. Loss of human decision-making
3. Making humans lazy

Although many other concerns about AI exist in education, these three are the most common and challenging in the current era. Additionally, no researcher can broaden the study beyond the scope.

Theoretical discussion

AI in education. Technology has impacted almost every sector; reasonably, it also needs time (Leeming, 2021). From telecommunication to communication and health to education, it plays a significant role and assists humanity in one way or another (Stahl A., 2021a, 2021b). No one can deny its importance and applications for life, which provides a solid reason for its existence and development. One of the most critical technologies is artificial intelligence (AI) (Ross, 2021). AI has applications in many sectors, and education is one. Many AI applications in education include tutoring, educational assistance, feedback, social robots, admission, grading, analytics, trial and error, virtual reality, etc. (Tahiru, 2021).

AI is based on computer programming or computational approaches; questions can be raised on the process of data analysis, interpretation, sharing, and processing (Holmes et al., 2019) and how the biases should be prevented, which may impact the rights of students as it is believed that design biases may increase with time and how it will address concerns associated with gender, race, age, income inequality, social status, etc. (Tarran, 2018). Like any other technology, there are also some challenges related to AI and its application in education and learning. This paper focuses on the ethical concerns of AI in education. Some problems are related to privacy, data access, right and wrong responsibility, and student records, to name a few (Petousi and Sifaki, 2020). In addition, data hacking and manipulation can challenge personal privacy and control; a need exists to understand the ethical guidelines clearly (Fjelland, 2020).

Perhaps the most important ethical guidelines for developing educational AI systems are well-being, ensuring workplace safety, trustworthiness, fairness, honoring intellectual property rights, privacy, and confidentiality. In addition, the following ten principles were also framed (Aiken and Epstein, 2000).

1. Ensure encouragement of the user.
2. Ensure safe human–machine interaction and collaborative learning
3. Positive character traits are to be ensured.
4. Overloading of information to be avoided
5. Build an encouraging and curious learning environment
6. Ergonomics features to be considered
7. Ensure the system promotes the roles and skills of a teacher and never replaces him
8. Having respect for cultural values
9. Ensure diversity accommodation of students
10. Avoid glorifying the system and weakening the human role and potential for growth and learning.

If the above principles are discussed individually, many questions arise while using AI technology in education. From its design and planning to use and impact, at every stage, ethical concerns arise and are there. It's not the purpose for which AI technology is developed and designed. Technology is advantageous for one thing but dangerous for another, and the problem is how to disintegrate the two (Vincent and van, 2022).

In addition to the proper framework and principles not being followed during the planning and development of AI for Education, bias, overconfidence, wrong estimates, etc., are additional sources of ethical concerns.

Security and privacy issues. Stephen Hawking once said that success in creating AI would be the most significant event in human history. Unfortunately, it might also be the last, unless we learn to avoid the risks. Security is one of the major concerns associated with AI and learning (Köbis and Mehner, 2021). Trustworthy artificial intelligence (AI) in education: Promises and challenges (Petousi and Sifaki, 2020; Owoc et al., 2021). Most educational institutions nowadays use AI technology in the learning process, and the area attracted research and interests. Many researchers agree that AI significantly contributes to e-learning and education (Nawaz et al. 2020; Ahmed and Nashat, 2020). Their claim is practically proved by the recent COVID-19 pandemic (Torda, 2020; Cirus et al., 2021). But AI or machine learning also brought many concerns and challenges to the education sector, and security and privacy are the biggest.

No one can deny that AI systems and applications are becoming a part of classrooms and education in one form or another (Sayantani, 2021). Each tool works according to its way, and the student and teacher use it accordingly. It creates an immersive learning experience using voices to access information and invites potential privacy and security risks (Gocen and Aydemir, 2020). While answering a question related to privacy concerns focuses on student safety as the number one concern of AI devices and usage. The same may go for the teacher's case as well.

Additionally, teachers know less about the rights, acts, and laws of privacy and security, their impact and consequences, and any violations cost to the students, teachers, and country (Vadapalli, 2021). Machine learning or AI systems are purely based on data availability. Without data, it is nothing, and the risk is unavoidable of its misuse and leaks for a lousy purpose (Hübner, 2021).

AI systems collect and use enormous data for making predictions and patterns; there is a chance of biases and discrimination (Weyerer and Langer, 2019). Many people are now concerned with the ethical attributes of AI systems and believe that the security issue must be considered in AI system development and deployment (Samtani et al., 2021). The Facebook-Cambridge Analytica scandal is one of the significant examples of how data collected through technology is vulnerable to privacy concerns. Although much work has been done, as the National Science Foundation recognizes, much more is still necessary (Calif, 2021). According to Kurt Markley, schools, colleges, and universities have big banks of student records comprising data related to their health, social security numbers, payment information, etc., and are at risk. It is necessary that learning institutions continuously re-evaluate and re-design the security practices to make the data secure and prevent any data breaches. The trouble is even more in remote learning environments or when information technology is effective (Chan and Morgan, 2019).

It is also of importance and concern that in the current era of advanced technology, AI systems are getting more interconnected to cybersecurity due to the advancement of hardware and software (Mengidis et al., 2019). This has raised significant concerns regarding the security of various stakeholders and emphasizes the procedures the policymakers must adopt to prevent or minimize the threat (ELever and Kifayat, 2020). It is also important to note that security concerns increase with network and endpoints in remote learning. One problem is that protecting e-learning technology from cyber-attacks is neither easy nor requires less money, especially in the education sector, with a limited budget for academic activities (Huls, 2021). Another reason why severe threat exists is because of very few technical staff in an educational institution; hiring them is another economic issue. Although, to some extent, using intelligent technology of AI and machine learning, the level and threat of security decrease, again, the issue is that neither every teacher is a professional and trained enough to use the technology nor able to handle the common threats. And as the use of AI in education increases, the danger of security concerns also increases (Taddeo et al., 2019). No one can run from the threat of AI concerning cybersecurity, and it behaves like a double-edged sword (Siau and Wang, 2020).

Digital security is the most significant risk and ethical concern of using AI in education systems, where criminals hack machines and sell data for other purposes (Venema, 2021). We alter our safety and privacy (Sutton et al., 2018). The question remains: whether our privacy is secured, and when will AI systems become able to keep our confidentiality connected? The answer is beyond human knowledge (Kirn, 2007).

Human interactions with AI are increasing day by day. For example, various AI applications, like robots, chatbots, etc., are used in e-learning and education. Many will learn human-like habits one day, but some human attributes, like self-awareness, consciousness, etc., will remain a dream. AI still needs data and uses it for learning patterns and making decisions; privacy will always remain an issue (Mhlanga, 2021). On the one hand, it is a fact that AI systems are associated with various human rights issues, which can be evaluated from case to case. AI has many complex pre-existing impacts regarding human rights because it is not installed or implemented against a blank slate but as a backdrop of societal conditions. Among many human rights that international law assures, privacy is impacted by it (Levin, 2018). From the discussed review, we draw the following hypothesis.

H_1: There is a significant impact of artificial intelligence on the security and privacy issues

Making humans lazy. AI is a technology that significantly impacts Industry 4.0, transforming almost every aspect of human

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life and society (Jones, 2014). The rising role of AI in organizations and individuals feared the persons like Elon Musk and Stephen Hawking. Who thinks it is possible when AI reaches its advanced level, there is a risk it might be out of control for humans (Clark et al., 2018). It is alarming that research increased eight times compared to the other sectors. Most firms and countries invest in capturing and growing AI technologies, skills, and education (Oh et al., 2017). Yet the primary concern of AI adoption is that it complicates the role of AI in sustainable value creation and minimizes human control (Noema, 2021).

When the usage and dependency of AI are increased, this will automatically limit the human brain's thinking capacity. This, as a result, rapidly decreases the thinking capacity of humans. This removes intelligence capacities from humans and makes them more artificial. In addition, so much interaction with technology has pushed us to think like algorithms without understanding (Sarwat, 2018). Another issue is the human dependency on AI technology in almost every walk of life. Undoubtedly, it has improved living standards and made life easier, but it has impacted human life miserably and made humans impatient and lazy (Krakauer, 2016). It will slowly and gradually starve the human brain of thoughtfulness and mental efforts as it gets deep into each activity, like planning and organizing. High-level reliance on AI may degrade professional skills and generate stress when physical or brain measures are needed (Gocen and Aydemir, 2020).

AI is minimizing our autonomous role, replacing our choices with its choices, and making us lazy in various walks of life (Danaher, 2018). It is argued that AI undermines human autonomy and responsibilities, leading to a knock-out effect on happiness and fulfilment (C. Eric, 2019). The impact will not remain on a specific group of people or area but will also encompass the education sector. Teachers and students will use AI applications while doing a task/assignment, or their work might be performed automatically. Progressively, getting an addiction to AI use will lead to laziness and a problematic situation in the future. To summarize the review, the following hypothesis is made:

H_2: There is a significant impact of artificial intelligence on human laziness

Loss of human decision-making. Technology plays an essential role in decision-making. It helps humans use information and knowledge properly to make stable decisions for their organization and innovations (Ahmad, 2019). Humans are producing large volumes of data, and to make it efficient, firms are adopting and using AI and kicking humans out of using the data. Humans think they are getting benefits and saving time by using AI in their decisions. But it is overcoming the human biological processes through having cognition capabilities (Jarrahi, 2018).

It is a fact that AI technologies and applications have many benefits. But technologies have severe negative consequences, and the limitation of their role in human decision-making is one of them. Slowly and gradually, AI limits and replaces the human role in decision-making. Human mental capabilities like intuitive analysis, critical thinking, and creative problem-solving are getting out of decision-making (Ghosh et al., 2019). Consequently, this will lead to their loss as there is a saying, use it or lose it. The speed of adaptation of AI technology is evident from the usage of AI in the strategic decision-making processes, which has increased from 10 to 80% in five years (Sebastian and Sebastian, 2021).

Walmart and Amazon have integrated AI into their recruitment process and make decisions about their product. And it's

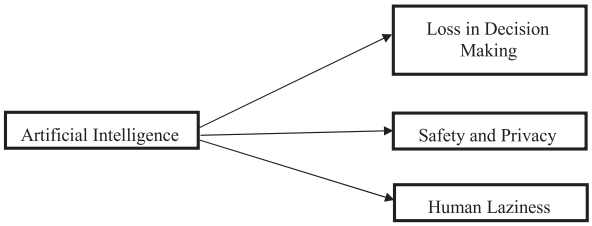

Fig. 1 Proposed model. The impact of artificial intelligence on human loss in decision making, laziness, and safety in education.

getting more into the top management decisions (Gilbert, 2017). Organizations use AI to analyze data and make complex decisions effectively to obtain a competitive advantage. Although AI is helping the decision-making process in various sectors, humans still have the last say in making any decision. This highlights the importance of humans' role in the process and the need to ensure that AI technology and humans work side by side (Meissner and Keding, 2021). The hybrid model of the human–machine collaboration approach is believed to merge in the future (Subramaniam, 2022).

The role of AI in decision-making in educational institutions is spreading daily. Universities are using AI in both academic and administrative activities. From students searching for program admission requirements to the issuance of degrees, they are now assisted by personalization, tutoring, quick responses, 24/7 access to learning and asking questions, and task automation are the leading roles AI plays in the education sector (Karandish, 2021).

In all the above roles, AI collects data, analyzes it, and then responds, i.e., makes decisions. It is necessary to ask some simple but essential questions: Does AI make ethical choices? The answer is AI was found to be racist, and its choice might not be ethical (Tilan, 2021). The second question is, does AI impact human decision-making capabilities? While using an intelligent system, applicants may submit their records directly to the designer and get approval for admission tests without human scrutiny. One reason is that the authorities will trust the system; the second may be the laziness created by task automation among the leaders.

Similarly, in keeping the records of students and analyzing their data, again, the choice will be dependent on the decision made by the system, either due to trust or due to the laziness created by task automation among the authorities. Almost in every task, the teachers and other workers lose the power of cognition while making academic or administrative decisions. And their dependency increases daily on the AI systems installed in the institution. To summarize the review, in any educational organization, AI makes operations automatic and minimizes staff participation in performing various tasks and making decisions. The teachers and other administrative staff are helpless in front of AI as the machines perform many of their functions. They are losing the skills of traditional tasks to be completed in an educational setting and consequently losing the reasoning capabilities of decision-making.

H_3: There is a significant impact of artificial intelligence with the loss of human decision making

Conceptual framework. Fig. 1

Methodology

Research design. The research philosophy focuses on the mechanism of beliefs and assumptions regarding knowledge development. It is precisely what the researcher works on while conducting research and mounting expertise in a particular area.

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In this research, the positivist philosophy of analysis is used. Positivism focuses on an observable social reality that produces the laws, just like generalizations. This philosophy uses the existing theory for hypotheses development in this study.

Furthermore, this philosophy is used because this study is about measurable and quantifiable data. The quantitative method is followed for data collection and analysis in this research. The quantitative practice focuses on quantifiable numbers and provides a systematic approach to assessing incidences and their associations. Moreover, while carrying out this study, the author evaluated the validity and reliability tools to ensure rigor in data. The primary approach is used because the data collected in this research is first-hand, which means it is collected directly from the respondents.

Sample and sampling techniques. The purposive sampling technique was used in this study for the primary data collection. This technique is used because it targets a small number of participants to participate in the survey, and their feedback shows the entire population (Davies and Hughes, 2014). Purposive sampling is a recognized non-probabilistic sampling technique because the author chose the participants based on the study's purpose. The respondents of this study were students at different universities in Pakistan and China. Following the ethical guidelines, consent was taken from the participants. After that, they were asked to give their responses through a questionnaire. The number of participants who took part in the study was 285. This data collection was around two months, from 4 July 2022 to 31 August 2022.

Measures. The survey instrument is divided into two parts. The initial portion of the questionnaire comprised demographic questions that included gender, age, country, and educational level. The second portion of the instrument had the Likert scale questions of the latent variables. This study model is composed of four latent variables. All four latent variables are measured through their developed Likert scale questions. All five measures of the latent variables are adopted from the different past studies that have developed and validated these scales. The measures of artificial intelligence are composed of seven items adopted from the study of Suh and Ahn (2022). The loss measures in decision-making consist of five items adopted from the study of Niese (2019). The measures of safety and security issues are composed of five items adopted from the study of Youn (2009). The measure of human laziness consists four items adopted from the study of Dautov (2020). All of them are measured on the Likert scale of five, 1 for the least level of agreement and five for the highest level of agreement. Table 1 shows the details of the items of each construct.

Common method bias. CMB is a major problem faced by the researcher working on the primary survey data research. There are many reasons for this dilemma. The primary reason is the response tendency, in which the respondents of the research rate equally to all questions (Jordan and Troth, 2020). A model's VIF values are not limited to multi-collinearity diagnostics but also indicate the common method bias (Kock, 2015). If the VIF values of the individual items present in the model are equal to or <3.3, then it is considered that the model is free from the common method bias. Table 2 shows that all the VIF values are <3.3, which indicates that the data collected by the primary survey is almost free from the issues of common bias.

Reliability and validity of the data. Reliability and validity confirm the health of the instrument and survey data for further analysis. Two tools are used in structural equation modeling for reliability: item reliability and construct reliability. The outer loading of each item gauges the item's reliability. Its threshold value is 0.706, but in some cases, even 0.5 is also acceptable if the basic assumption of the convergent validity is not violated (Hair and Alamer, 2022). Cronbach's Alpha and composite reliability are the most used tools to measure construct reliability. The threshold value is 0.7 (Hair Jr et al., 2021). Table 3 shows that all the items of each construct have outer loading values greater than 0.7. Only one item of artificial intelligence and one item of decision making is below 0.7 but within the minimum limit of 0.4, and both AVE values are also good. While each construct Cronbach's alpha and composite reliability values are >0.7, both measures of reliability, item reliability, and construct reliability are established. For the validity of the data there are also two measures used one is convergent validity, and the other is discriminant validity. For convergent validity, AVE values are used. The threshold value for the AVE is 0.5 (Hair and Alamer, 2022). From the table of reliability and validity, all the constructs have AVE values >0.5, indicating that all the constructs are convergently valid.

In Smart-PLS, three tools are used to measure discriminant validity: the Farnell Larker criteria, HTMT ratios, and the cross-loadings of the items. The threshold value for the Farnell Licker criteria is that the diagonal values of the table must be greater than the values of its corresponding rows and columns. Table 4 shows that all the diagonal values of the square root of the AVE are greater than their corresponding values of both columns and rows. The threshold value for the HTMT values is 0.85 or less (Joe F. Hair Jr et al., 2020). Table 5 shows that all the values are less than 0.85. Table 6 shows that they must have self-loading with their construct values greater than the cross-loading with other constructs. Table 6 shows that all the self-loadings are greater than the cross-loadings. All three above measures of discriminant validity show that the data is discriminately valid.

Results and discussion

Demographic profile of the respondents. Table 7 shows the demographic characteristics of the respondents. Among 285 respondents, 164 (75.5%) are male, while 121 (42.5%) are female. The data was collected from different universities in China and Pakistan. The table shows that 142 (50.2%) are Chinese students, and 141 (49.8%) are Pakistani students. The age group section shows that the students are divided into three age groups, <20 years, 20–25 years, and 26 years and above. Most students belong to the age group 20–25 years, which is 140 (49.1%), while 26 (9.1%) are <20 years old and 119 (41.8%) are 26 years and above. The fourth and last section of the table shows the program of the student's studies. According to this, 149 (52.3%) students are undergraduates, 119 (41.8%) are graduates, and 17 (6%) are post-graduates.

Structural model. The structural model explains the relationships among study variables. The proposed structural model is exhibited in Fig. 2.

Regression analysis. Table 8 shows the total direct relationships in the model. The first direct relationship is between artificial intelligence to loss in human decision-making, with a beta value of 0.277. The beta value shows that one unit increase in artificial intelligence will lose human decision-making by 0.277 units among university students in Pakistan and China. This relationship having the t value of 5.040, greater than the threshold value of 1.96, and a p-value of 0.000, <0.05, shows that the relationship is statistically significant. The second relationship is between artificial intelligence the human

Table 1 Measures.

Construct	Codes	Items
Artificial intelligence	AI1	It is interesting to use AI.
	AI2	AI could make everything better.
	AI3	AI is very important for developing society
	AI4	AI is necessary for everyone
	AI5	AI produces more good than bad.
	AI6	I think AI makes life more convenient.
	AI7	AI helps me solve problems in real life.
Decision making	DM1	How easy or difficult was the PROCESS of trying to find an answer?
	DM2	I believe there is a good match between my decision and the decision support technology.
	DM3	I believe the decision support technology is not well suited for my decision.
	DM4	I believe there is an excellent fit between my decision and the decision support technology.
	DM5	I believe there is a mismatch between the decision I've made and the decision to support technology
Human laziness	HL1	Seeing what to do but don't want to do it
	HL2	Postponing what should be done until the end
	HL3	I avoid more complex jobs, affairs or assignments
	HL4	Put aside work/homework and do what you like to do first. (For example, play game first, then do business)
Security and privacy issues	SP1	By using AI, I am experiencing financial loss
	SP2	By using AI, I am Experiencing identity theft
	SP3	I am concerned about how companies collect and use personal information online.
	SP4	I always received junk emails or unwanted mail
	SP5	I am experiencing a feeling that my personal information may be misused

Table 2 Multicollinearity statistics.

Constructs	Items	VIF values
Artificial intelligence	A1	2.019
	A2	2.069
	A3	2.113
	A4	1.813
	A5	1.940
	A6	1.641
	A7	2.021
Decision making	DM1	1.394
	DM2	1.12[illegible]
	DM3	1.[illegible]51
	DM4	1.701
Human laziness	HL1	2.040
	HL2	[illegible]066
	HL3	2.397
	HL4	[illegible]109
Safety & privacy	SP1	1.514
	SP[illegible]	1.729
	SP[illegible]	1.612

Above are the individual VIF values of the indicator items, along with their constructs.

Table 3 Reliability and validity.

Constructs	Items	Loadings	Cronbach alpha	Composite reliability	AVE
Artificial intelligence	A1	0.772	0.873	0.902	0.569
	A2	0.782			
	A3	0.792			
	A4	0.746			
	A5	0.777			
	A6	0.646			
	A7	0.757			
Decision making	DM1	0.802	0.715	0.818	0.540
	DM2	0.454			
	DM3	0.855			
	DM4	0.762			
Safety & privacy	SP1	0.818	0.776	0.870	0.691
	SP3	0.838			
	SP4	0.837			
Human laziness	HL1	0.842	0.872	0.912	0.722
	HL2	0.839			
	HL3	0.870			
	HL4	0.848			

Items DM5, SP2, and SP5 were removed from the model due to low outer loading values.

laziness. The beta value for this relationship is 0.689, which shows that one unit increase in artificial intelligence will make the students of Pakistan and China universities lazy by 0.689 units. The *t*-value for the relationship is 2.257, which is greater than the threshold value of 1.96, and a *p*-value of 0.000, which is smaller than the threshold value of 0.05, which shows that this relationship is also statistically significant. The third and last relationship is from artificial intelligence to security and privacy issues of Pakistani and Chinese university students. The beta value for this relationship is 0.686, which shows that a one-unit increase in artificial intelligence will increase security and privacy issues by 0.686. The *t*-value for the relationship is 17.105, which is greater than the threshold value of 1.96, and the *p*-value is 0.000, which is smaller than a threshold value of 0.05, indicating that this relationship is also statistically significant.

Hypothesis testing. Table 8 also indicates that the results support all three hypotheses.

Model fitness. Once the reliability and validity of the measurement model are confirmed, the structural model fitness must be assessed in the next step. For the model fitness, several measures are available in the SmartPLS, like SRMR, Chi-square, NFI, etc., but most of the researcher recommends the SRMR for the model fitness in the PLS-SEM. When applying PLS-SEM, a value <0.08 is generally considered a good fit (Hu and Bentler, 1998). However, the table of model fitness shows that the SRMR value is 0.06, which is less than the threshold value of 0.08, which indicates that the model is fit.

Predictive relevance of the model. Table 9 shows the model's prediction power, as we know that the model has total dependent variables. Then there are three predictive values for the model for each variable. The threshold value for predicting the model power is greater than zero. However, Q^2 values of 0.02, 0.15, and 0.35,

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Table 4 Fornell Larcker criteria.

	Artificial intelligence	Decision making	Human laziness	Safety & privacy
Artificial intelligence	**0.755**			
Decision making	0.277	**0.735**		
Human laziness	0.689	0.288	**0.85**	
Safety & privacy	0.686	0.241	0.492	**0.831**

The diagonal values are the square root of the AVEs.

Table 5 HTMT values.

	Artificial intelligence	Decision making	Human laziness
Decision making	0.311		
Human laziness	0.787	0.338	
Safety & privacy	0.831	0.309	0.596

Table 6 Cross-loadings.

	Artificial intelligence	Decision making	Human laziness	Safety & privacy
A1	**0.772**	0.263	0.533	0.532
A2	**0.782**	0.202	0.565	0.474
A3	**0.792**	0.254	0.613	0.487
A4	**0.746**	0.212	0.505	0.506
A5	**0.777**	0.278	0.496	0.601
A6	**0.646**	0.082	0.43	0.475
A7	**0.757**	0.141	0.482	0.545
DM1	0.247	**0.802**	0.224	0.18
DM2	0.082	**0.454**	0.093	0.08
DM3	0.263	**0.855**	0.289	0.23
DM4	0.144	**0.762**	0.176	0.179
HL1	0.586	0.189	**0.842**	0.398
HL2	0.566	0.301	**0.839**	0.411
HL3	0.581	0.297	**0.870**	0.4[illegible]
HL4	0.606	0.196	**0.84[illegible]**	0.438
SP1	0.582	0.132	0.38[illegible]	**0.818**
SP3	0.53	0.268	0.37[illegible]	**0.838**
SP4	0.594	0.207	0.463	**0.837**

All the bold numbers are self-loading while non-bold numbers are cross-loadings of each construct.

Table 7 Demographic distribution of respondents.

	No.	Percentage
Gender		
Male	164	57.5
Female	121	42.5
Total	285	100
Country		
China	142	49.8
Pakistan	143	50.2
Total	285	100
Age group		
[illegible] years	26	9.1
20–25 years	140	49.1
26 years and above	119	41.8
Total	285	100
Program of study		
Undergraduate	149	52.3
Graduate	119	41.8
Post-Graduate	17	6.0
Total	285	100

The above is the demographic distribution of the data collected by students from different Pakistan and China universities.

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respectively, indicate that an independent variable of the model has a low, moderate or high predictive relevance for a certain endogenous construct (Sarstedt et al., 2013). Human laziness has the highest predictive relevance with a Q^2 value of 0.338, which shows a moderate effect. Safety and security issues have the second largest predictive relevance with the Q^2 value of 0.314, which also shows a moderate effect. The last and smallest predictive relevance in decision-making with a Q^2 value of 0.033 which shows a low effect. A greater Q^2 value shows that the variable of model has the highest prediction power.

Importance performance matrix analysis (IPMA). Table 10 shows the importance and performance of each independent variable for the dependent variables. We see that artificial intelligence has the same performance of 68.78% for all three variables: human laziness, decision-making, safety, and security. While the importance of artificial intelligence, human laziness is 68.9%, loss in decision-making is 25.1%, and safety and security are 74.6%. This table shows that safety and privacy have the highest importance, and their performance is recommended to be increased to meet the important requirements. Figures 3–5 also show all three variables' importance compared to performance with artificial intelligence.

Multi-group analysis (MGA). Multigroup analysis is a technique in structural equation modeling that compares the effects of two classes of categorical variables on the model's relationships. The first category is gender, composed of male and female subgroups or types. Table 10 shows the gender comparison for all three relationships. The data record shows that there were 164 males and 121 females. The p-values of all three relationships are >0.05, which shows that gender is not moderate in any of the relationships. Table 10 shows the country-wise comparison for all three relationships in the model. The p-values of all three relationships are >0.05, indicating no moderating effect of the country on all three relationships. The data records show 143 Pakistanis and 142 Chinese based on the country's origin.

Discussion. AI is becoming an increasingly important element of our lives, with its impact felt in various aspects of our daily life. Like any other technological advancement, there are both benefits

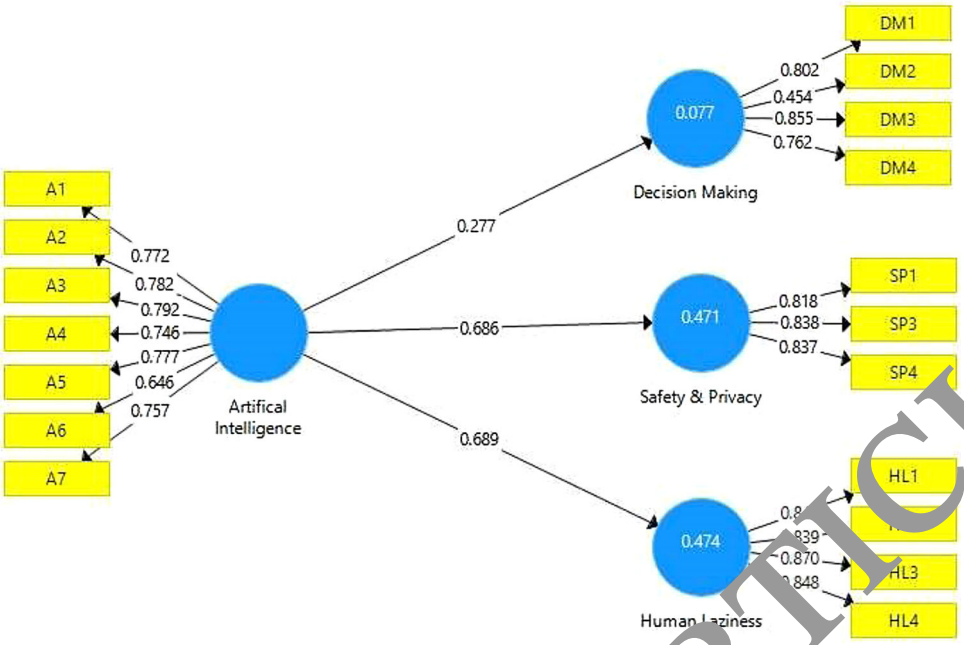

Fig. 2 Results model for the Impact of artificial intelligence on human loss in decision-making, laziness and safety in education.

and challenges. This study examined the association of AI with human loss in decision-making, laziness and safety and privacy concerns. The results given Tables 11 and 12 show that AI has a significant positive relationship with all these variables. The findings of this study also support that the use of AI technologies is creating problems for users related to security and privacy. Previous research has also shown similar results (Bartoletti, 2019; Saura et al., 2022; Bartneck et al., 2021). Using AI technology in an educational organization also leads to security and privacy issues for students, teachers, and institutions. In today's information age, security and privacy are critical concerns of AI technology use in educational organizations (Kamenskii, 2022). Skills specific to using AI technology are required for its effective use. Insufficient knowledge about the use will lead to security and privacy issues (Vazhayil and Shetty, 2019). Mostly educational firms do not have AI technology experts in managing it, which again increases its vulnerability in the context of security and privacy issues. Even if its users have sound skills and the firms have experienced AI managers, no one can deny that any security or privacy control could be a human mistake and could lead to serious security and privacy problems. Moreover, the fact that people with different levels of skills and competence interact in educational organizations also leads to the hacking or leaking of personal and institutional data (Kamenskih, 2022). AI is based on algorithms and uses large data sets to automate instruction (Araujo et al., 2020). Any mistake in the algorithms will create serious problems, and unlike humans, it will repeat the same mistake in making its own decisions. It also increases the threat to institutional and student data security and privacy. The same challenge is coming from the student end. They can be easily victimized as they are not excellently trained to use AI (Asaro, 2019). With the increase in the number of users, competence division and distance, safety and privacy concerns increase (Lv and Singh, 2020). The consequences depend upon the nature of the attack and the data been leaked or used by the attackers (Vassileva, 2008).

The findings show that AI-based products and services are increasing the human laziness factor among those relying more on AI. However, there were not too many studies conducted on this factor by the researcher in the past, but the numerous researches available in the literature also endorse the findings of this study (Fan, 2022; Bartoletti, 2019). AI in education leads to the creation of laziness in humans. AI performs repetitive tasks in an automated manner and does not let humans memorize, use analytical mind skills, or use cognition (Nikita, 2023). It leads to an addiction behavior not to use human capabilities, thus making humans lazy. Teachers and students who use AI technology will slowly and gradually lose interest in doing tasks themselves. This is another important concern of AI in the education sector (Crispin Andrews). The teachers and students are getting lazy and losing their decision-making abilities as much of the work is assisted or replaced by AI technology (BARON, 2023). Posner and Fei-Fei (2020) suggested it is time to change AI for education.

The findings also show that the access use of AI will gradually lead to the loss of human decision-making power. The results also endorsed the statement that AI is one of the major causes of the human loss of decision-making power. Several researchers from the past have also found that AI is a major cause responsible for the gradual loss of people's decision-making (Pomerol, 1997; Duan et al., 2019; Cukurova et al., 2019). AI performs repetitive tasks in an automated manner and does not let humans memorize, use analytical mind skills, or use cognition, leading to the loss of decision-making capabilities (Nikita, 2023). An online environment for education can be a good option (VanLangen, 2021), but the classroom's physical environment is the prioritized education mode (Dib and Adamo, 2014). In a real environment, there is a significant level of interaction between the teacher and students, which develop the character and civic bases of the students, e.g., students can learn from other students, ask teachers questions, and even feel the education environment. Along with the curriculum, they can learn and adopt many positive understandings (Quinlan et al., 2014). They can learn to use their cognitive power to choose options, etc. But unfortunately, the use of AI technology minimizes the real-time physical interaction (Mantello et al., 2021) and the education environment between students and teachers, which has a considerable impact on students' schooling, character, civic responsibility, and their power to make decisions, i.e., use their cognition. AI technology reduces the cognitive power of humans who make their own decisions (Hassani and Unger, 2020).

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Table 8 Regression analysis.

Relationships	β	Mean	STDEV	*t* values	*P*-values	Remarks
Artificial intelligence → Decision making	0.277	0.287	0.055	5.040	0.000	Supported
Artificial intelligence → Human laziness	0.689	0.690	0.030	23.257	0.000	Supported
Artificial intelligence → Safety & privacy	0.686	0.684	0.040	17.105	0.000	Supported

All three relationships in this table are based on the hypothesis of this study and all are statistically significant.

Table 9 IPMA analysis.

	Importance	Performances
Decision making		
Artificial intelligence	0.251	68.78
Human laziness		
Artificial intelligence	0.689	68.78
Safety and security		
Artificial intelligence	0.746	68.78

Table 10 Multi-group (analysis of gender)[a].

	β-diff (male–female)	*p*-value new (male vs. female)
Artificial intelligence → Decision making	−0.019	0.875
Artificial intelligence → Human laziness	0.077	0.194
Artificial intelligence → Safety & privacy	0.032	0.670

Multi group analysis (country wise)[b]

	β-diff (China– Pakistan)	*p*-Value new (China vs. Pakistan)
Artificial intelligence → Decision making	0.133	0.188
Artificial intelligence → Human laziness	−0.017	0.776
Artificial intelligence → Safety & privacy	−0.034	0.656

[a]Among 285 respondents 164 were males and 121 were females.
[b]Among 285 respondents 142 were Chinese and 143 were Pakistani.

Table 11 Model fitness.

	Saturated model	Estimated model
SRMR	0.065	0.065
d_ULS	0.73	0.793
d_G	0.281	0.286
Chi-Square	468.35	473.968
NFI	0.811	0.809

Table 12 Predictive relevance of the model.

	SSO	SSE	Q^2 (=1 − SSE/SSO)
Artificial intelligence	1995	1995	
Decision making	1140	1101.942	0.033
Human laziness	1140	754.728	0.338
Safety & privacy	855	586.125	0.314

Q^2 values show the prediction power of the model.

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AI technology has undoubtedly transformed or at least affected many fields (IEEE, 2019; Al-Ansi and Al-Ansi, 2023). Its applications have been developed for the benefit of humankind (Justin and Mizuko, 2017). As technology assists employees in many ways, they must be aware of the pros and cons of the technology and must know its applications in a particular field (Nadir et al., 2022). Technology and humans are closely connected; the success of one is strongly dependent on the other; therefore, there is a need to ensure the acceptance of technology for human welfare (Ho et al., 2022). Many researchers have discussed the user's perception of a technology (Vazhayil and Shetty, 2019), and many have emphasized its legislative and regulatory issues (Khan et al., 2014). Therefore, careful selection is necessary to adopt or implement any technology (Ahmad and Shahid, 2015). Once imagined in films, AI now runs a significant portion of the technology, i.e., health, transport, space, and business. As AI enters the education sector, it has been affected to a greater extent (Hübner, 2021). AI further strengthened its role in education, especially during the recent COVID-19 pandemic, and invaded the traditional way of teaching by providing many opportunities to educational institutions, teachers, and students to continue their educational processes (Štrbo, 2020; Al-Ansi, 2022; Akram et al., 2021). AI applications/technology like chatbots, virtual reality, personalized learning systems, social robots, tutoring systems, etc., assist the educational environment in facing modern-day challenges and shape education and learning processes (Schiff, 2021). In addition, it is also helping with administrative tasks like admission, grading, curriculum setting, and record-keeping, to name a few (Andreotta and Kirkham, 2021). It can be said that AI is likely to affect, enter and shape the educational process on both the institutional and student sides to a greater extent (Xie et al., 2021). This phenomenon hosts some questions regarding the ethical concerns of AI technology, its implementation, and its impact on universities, teachers, and students.

The study has similar findings to the report published by the Harvard Kennedy School, where AI concerns like privacy, automation of tasks, and decision-making are discussed. It says that AI is not the solution to government problems but helps enhance efficiency. It is important to note that the report does not deny the role of AI but highlights the issues. Another study says that AI-based and human decisions must be combined for more effective decisions. i.e., the decisions made by AI must be evaluated and checked, and the best will be chosen by humans from the ones recommended by AI (Shrestha et al., 2019). The role of AI cannot be ignored in today's technological world. It assists humans in performing complex tasks, providing solutions to many complex problems, assisting in decision-making, etc. But on the other hand, it is replacing humans, automating tasks, etc., which creates challenges and demands for a solution (Duan et al., 2019). People are generally concerned about risks and have conflicting opinions about the fairness and effectiveness of AI decision-making, with broad perspectives altered by individual traits (Araujo et al. 2020).

There may be many reasons for these controversial findings, but the cultural factor was considered one of the main factors (Elliott, 2019). According to researchers, people with high cultural values have not adopted the AI problem, so this cultural constraint remains a barrier for the AI to influence their behaviors (Di Vaio et al., 2020; Mantelero, 2018). The other

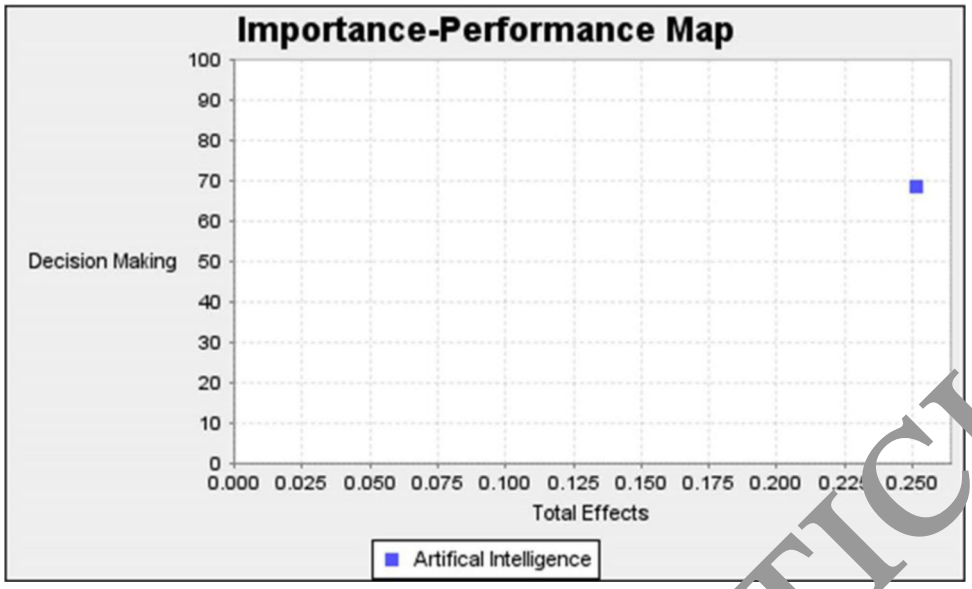

Fig. 3 Importance-performance map—human loss in decision making and artificial intelligence.

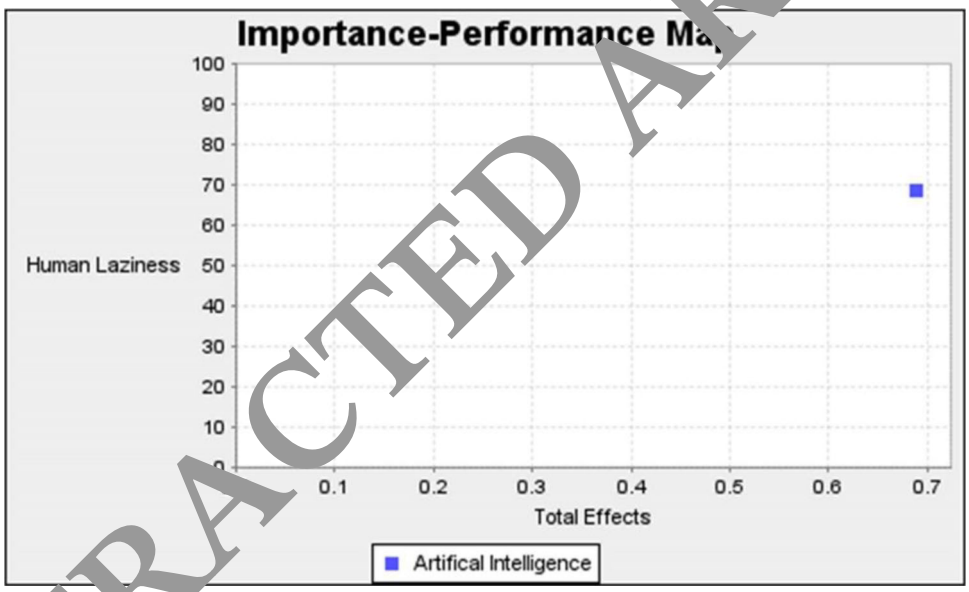

Fig. 4 Importance-performance map—human laziness and artificial intelligence.

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thing is that privacy is a term that has a different meaning from culture to culture (Ho et al., 2022). In some cultures, people consider minimal interference in personal life a big privacy issue, while in some cultures, people even ignore these types of things (Marullo et al., 2021). The results are similar to Zhang et al. (2022), Friken and Epstein (2000), and Bhbosale et al. (2020), which focus on the ethical issues of AI in education. These studies show that AI use in education is the reason for laziness among students and teachers. In short, the researchers are divided on the AI concerns in education, just like in other sectors. But they agree on the positive role AI plays in education. AI in education leads to laziness, loss of decision-making capabilities, and security or privacy issues. But all these issues can be minimized if AI is properly implemented, managed, and used in education.

Implications. The research has important implications for technology developers, the organization that adopts the technology, and the policymakers. The study highlights the importance of addressing ethical concerns during AI technology's development and implementation stage. It also provides guidelines for the government and policymakers regarding the issues arising with AI technology and its implementation in any organization, especially in education. AI can revolutionize the education sector, but it has some potential drawbacks. Implications suggest that we must be aware of the possible impact of AI on laziness, decision-making, privacy, and security and that we should design AI systems that have a very minimal impact.

Managerial Implications: Those associated with the development and use of AI technology in education need to find out the advantages and challenges of AI in this sector and balance these advantages with the challenges of laziness, decision-making, and privacy or security while protecting human creativity and intuition. AI systems should be designed to be transparent and ethical in all

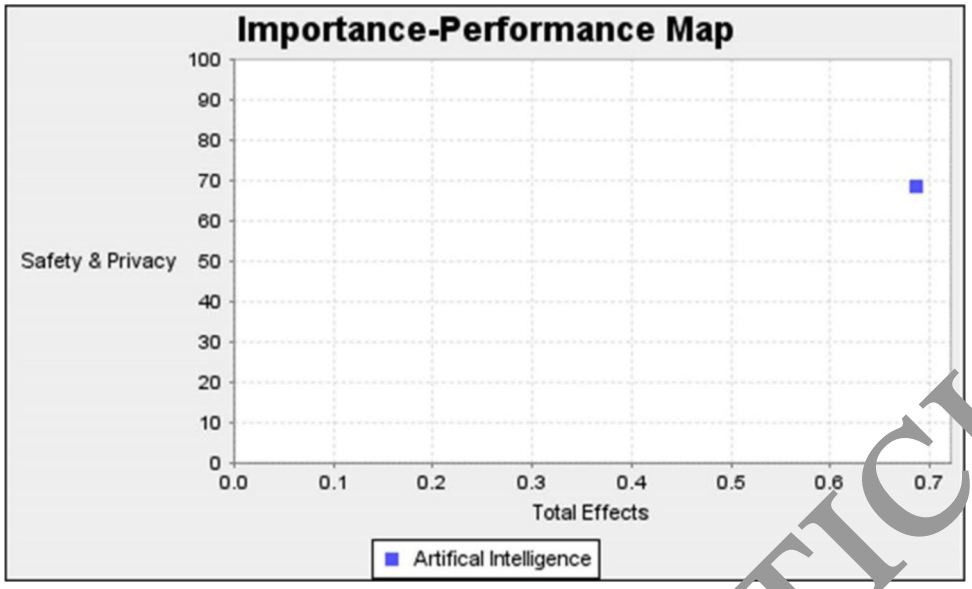

Fig. 5 Importance-performance map—safety and privacy and artificial intelligence.

manners. Educational organizations should use AI technology to assist their teachers in their routine activities, not to replace them.

Theoretical Implications: A loss of human decision-making capacity is one of the implications of AI in education. Since AI systems are capable of processing enormous amounts of data and producing precise predictions, there is a risk that humans would become overly dependent on AI in making decisions. This may reduce critical thinking and innovation for both students and teachers, which could lower the standard of education. Educators should be aware of how AI influences decision-making processes and must balance the benefits of AI with human intuition and creativity. AI may potentially affect school security. AI systems can track student behavior, identify potential dangers, and identify situations where children might require more help. There are worries that AI could be applied to unjustly target particular student groups or violate students' privacy. Therefore, educators must be aware of the potential ethical implications of AI and design AI systems that prioritize security and privacy for users and educational organizations. AI makes people lazier is another potential impact on education. Teachers and learners may become more dependent on AI systems and lose interest in performing activities or learning new skills or methodologies. This might lead to a decline in educational quality and a lack of personal development among people. Therefore, teachers must be aware of the possible detrimental impacts of AI on learners' motivation and should create educational environments that motivate them to participate actively in getting an education.

Conclusion

AI can significantly affect the education sector. Though it benefits education and assists in many academic and administrative tasks, its concerns about the loss of decision-making, laziness, and security may not be ignored. It supports decision-making, helps teachers and students perform various tasks, and automates many processes. Slowly and gradually, AI adoption and dependency in the education sector are increasing, which invites these challenges. The results show that using AI in education increases the loss of human decision-making capabilities, makes users lazy by performing and automating the work, and increases security and privacy issues.

Recommendations.

1. The teacher's foremost priority should be ensuring that AI will not cause any ethical concerns in education. Realistically, it is impossible, but at least severe ethical problems (both individual and societal) can be minimized during this phase.
2. AI technology and applications in education need to be backed by solid and secure algorithms that ensure the technology's security, privacy, and users.
3. Bias behavior of AI must be minimized, and issues of loss of human decision-making and laziness must be addressed.
4. Dependency on AI technology in decision-making must be reduced to a certain level to protect human cognition.
5. Teachers and students should be given training before using AI technology.

Future work.

1. Research can be conducted to study the other concerns of AI in education which were not studied.
2. Description and enumeration of the documents under analysis.
3. Procedure for the analysis of documents. Discourse analysis and categorization.
4. Similar studies can be conducted in other geographic areas and countries.

Limitations. This study is limited to three basic ethical concerns of AI: loss of decision-making, human laziness, and privacy and security. Several other ethical concerns need to be studied. Other research methodologies can be adopted to make it more general.

Data availability

The data set generated during and/or analyzed during the current study is submitted as supplementary file and can also be obtained from the corresponding author upon reasonable request.

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Author contributions

The authors confirm their contribution to the paper as follows: Introduction: SFA and HH; Materials: MMA and MKH, MI; Methods: MI and SFA; Data collection: SFA, MMA, MKR, MI, MMA, and AAM; Data analysis and interpretation SFA, HH, MI, and AAM; Draft preparation: MMA, MKR, MI, MMA, and AAM; Writing and review: SFA, HH, MMA, MKR, MI, MMA, and AAM. All authors read, edited, and finalized the manuscript.

Competing interests

The authors declare no competing interests.

Ethical approval

The evaluation survey questionnaire and methodology were examined, approved, and endorsed by the research ethics committee of the University of Gwadar on 1 March 2021 (see supplementary information). The study meets the requirements of the National Statement on Ethical Conduct in Human Research (2007). The procedures used in this study adhere to the tents of the declaration of Helsinki.

Informed consent

Informed consent was obtained from all participants before the data was collected. We informed each participant of their rights, the purpose of the study and to safeguard their personal information.

Additional information

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